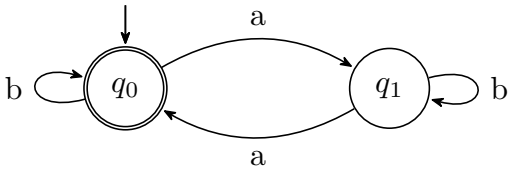
オートマトン・言語理論 レポート課題 01

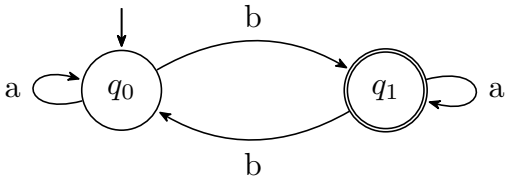
C5TB2006 安倍大陽

2026 年 10 月 19 日

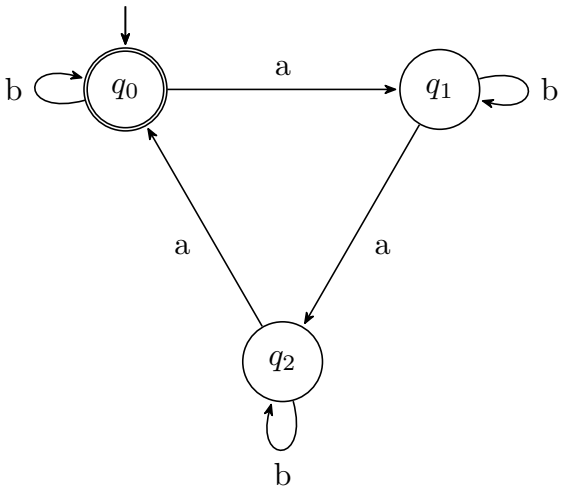
1 $L_1 = \{w \in \Sigma^* \mid \langle w \rangle_a = 2k, k \in \mathbb{N}\}$



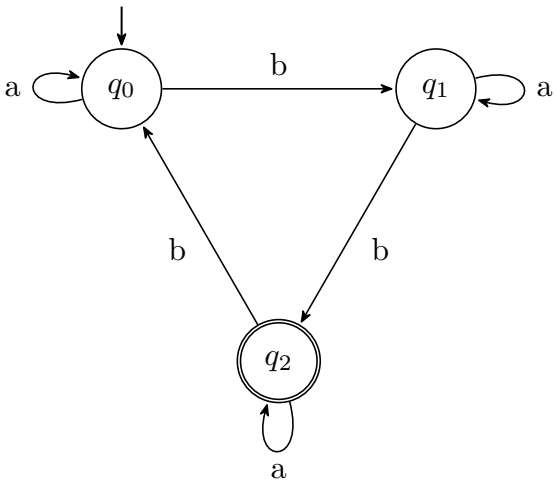
2 $L_2 = \{w \in \Sigma^* \mid \langle w \rangle_b = 2k + 1, k \in \mathbb{N}\}$



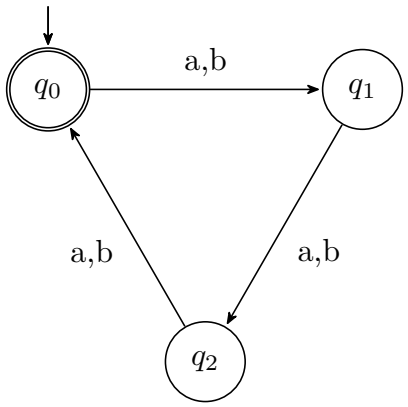
3 $L_3 = \{w \in \Sigma^* \mid \langle w \rangle_a = 3k, k \in \mathbb{N}\}$



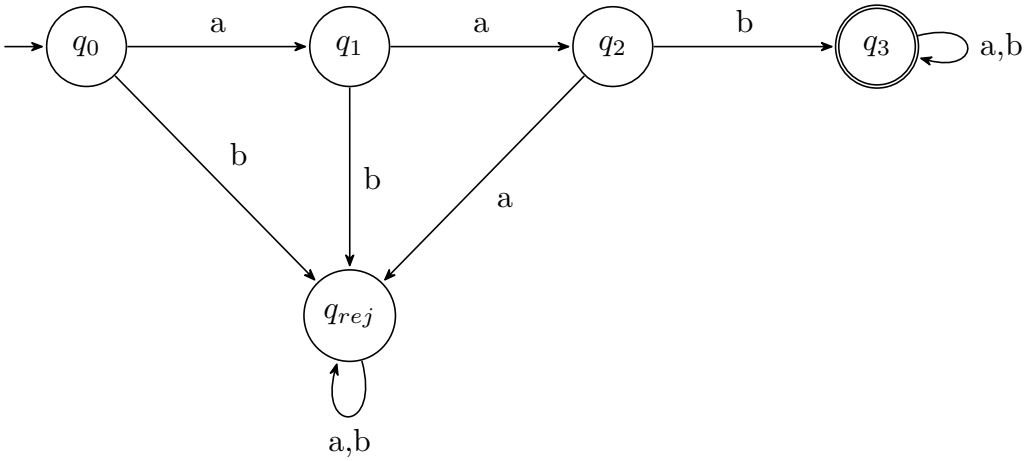
4 $L_4 = \{w \in \Sigma^* \mid \langle w \rangle_b = 3k + 2, k \in \mathbb{N}\}$



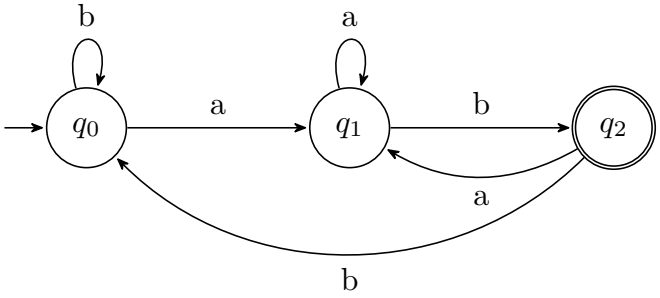
5 $L_5 = \{w \in \Sigma^* \mid |w| = 3k, k \in \mathbb{N}\}$



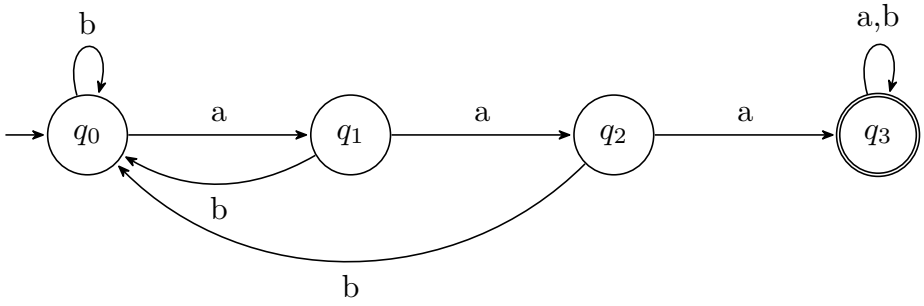
6 $L_6 = \{\mathsf{aab}v \mid v \in \Sigma^*\}$



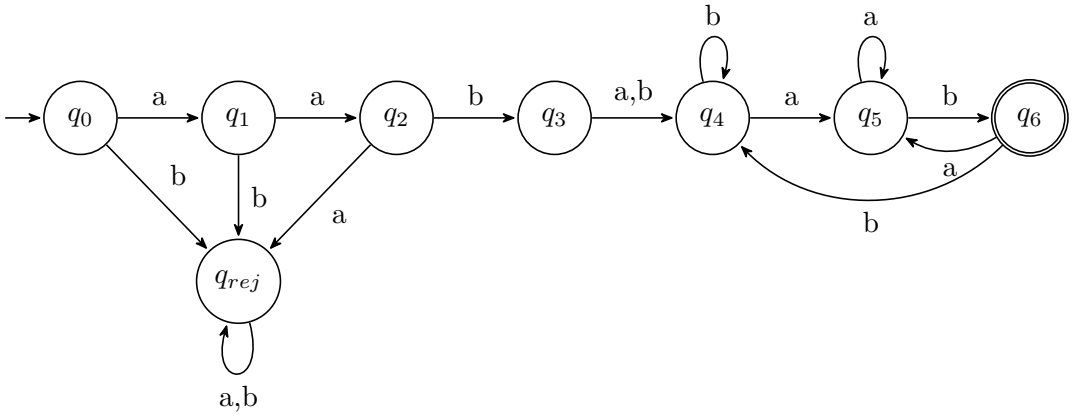
7 $L_7 = \{u\mathsf{ab} \mid u \in \Sigma^*\}$



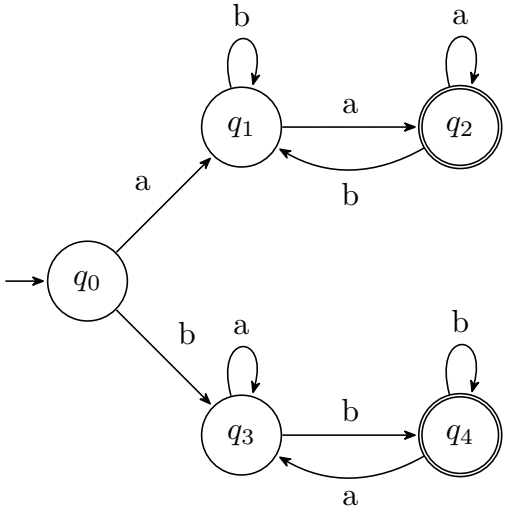
8 $L_8 = \{u\mathsf{aaa}v \mid u, v \in \Sigma^*\}$



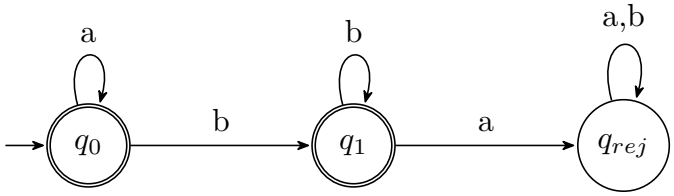
9 $L_9 = \{\mathsf{aab}v\mathsf{ab} \mid v \in \Sigma^*\}$



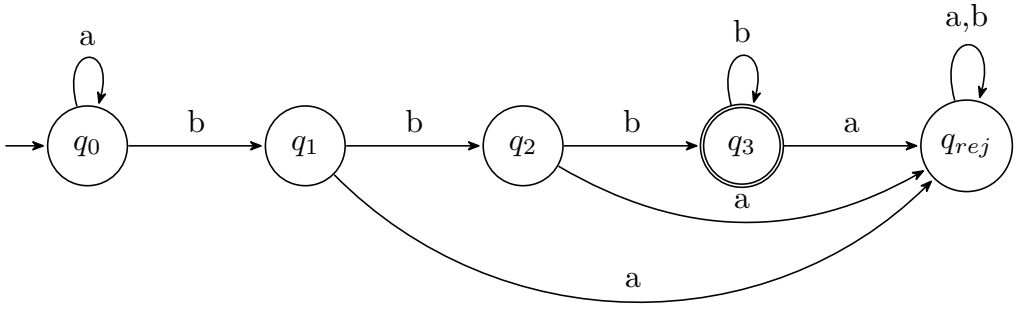
10 $L_{10} = \{\mathsf{a}v\mathsf{a} \mid v \in \Sigma^*\} \cup \{\mathsf{b}v\mathsf{b} \mid v \in \Sigma^*\}$



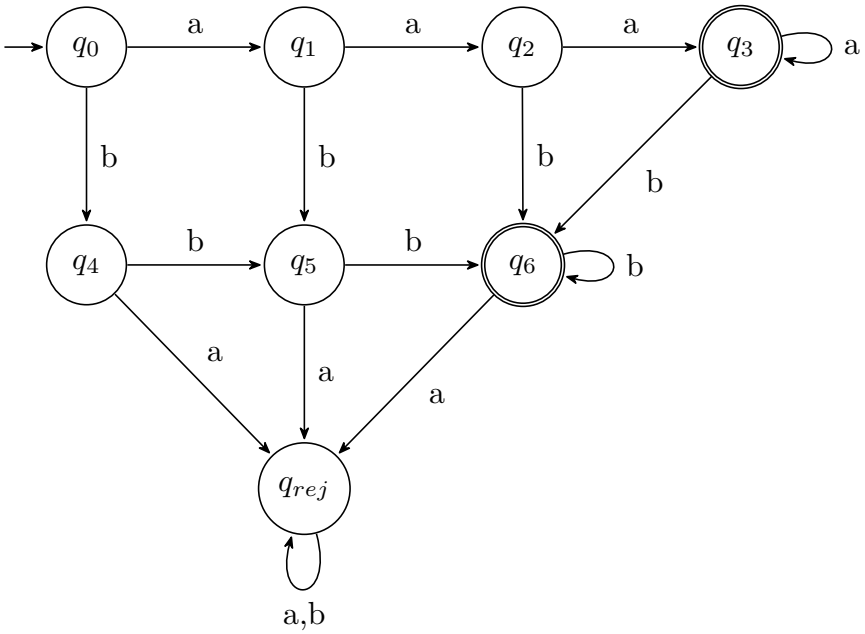
11 $L_{11} = \{\mathsf{a}^i\mathsf{b}^j \mid i \geq 0, j \geq 0\}$



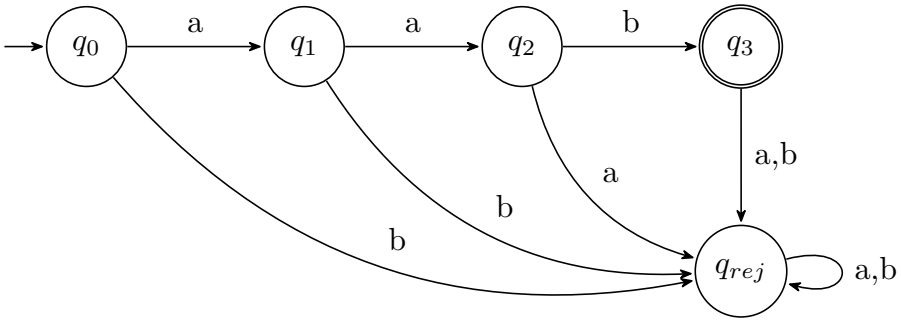
12 $L_{12} = \{\mathsf{a}^i\mathsf{b}^j \mid i \geq 0, j \geq 3\}$



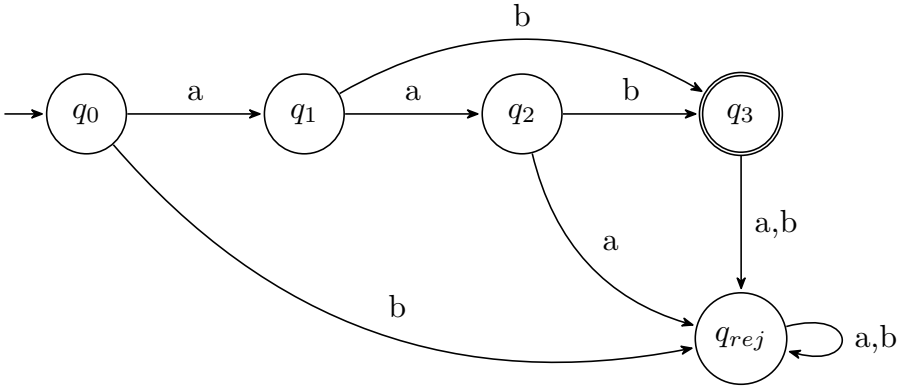
13 $L_{13} = \{\mathsf{a}^i\mathsf{b}^j \mid i + j \geq 3\}$



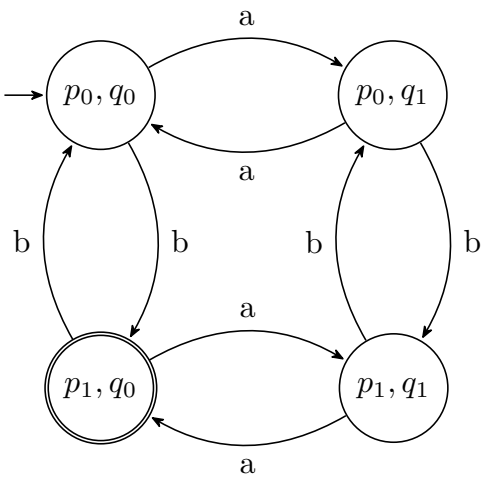
14 $L_{14} = \{\mathsf{aab}\}$



15 $L_{15} = \{\mathsf{aab}, \mathsf{ab}\}$



16 $L_{16} = L_1 \cap L_2$



17 $L_{17} = L_1 \cup L_2$

